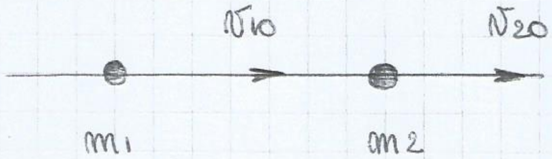
Unidad VIII: Teoremas de Conservación

Choque elástico (de partículas en una dimensión):

1) Elástico $\Rightarrow E_c = cte$

2) Plástico o inelástico

$$\left.\begin{matrix} 1) \\ 2) \end{matrix}\right\} \; P = cte = M\, v_{cm} \Rightarrow v_{cm} = cte$$



Sistema de laboratorio

$$P = m_1 v_{10} + m_2 v_{20}$$

Antes: $m_1 v_{10} + m_2 v_{20} = P = cte$

$$\frac{1}{2} m_1 v_{10}^2 + \frac{1}{2} m_2 v_{20}^2 = cte$$

Después: $m_1 v_1 + m_2 v_2 = P = cte$

$$\frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 = cte$$

$$\begin{cases} m_1 v_{10} + m_2 v_{20} = m_1 v_1 + m_2 v_2 & (1) \\ \frac{1}{2} m_1 v_{10}^2 + \frac{1}{2} m_2 v_{20}^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 & (2) \end{cases}$$

$$\begin{cases} m_1 (v_{10} - v_1) = m_2 (v_2 - v_{20}) & (1) \\ \frac{1}{2} m_1 (v_{10}^2 - v_1^2) = \frac{1}{2} m_2 (v_2^2 - v_{20}^2) & (2) \end{cases}$$

$$\begin{cases} m_1 (v_{10} - v_1) = m_2 (v_2 - v_{20}) & (1) \\ \frac{1}{2} m_1 (v_{10} - v_1)(v_{10} + v_1) = \frac{1}{2} m_2 (v_2 - v_{20})(v_2 + v_{20}) & (2) \end{cases}$$

(2) / (1) $\quad v_{10} + v_1 = v_2 + v_{20}$

$$v_{10} - v_{20} = v_2 - v_1$$

$$v_2 = v_{20}\left(\frac{m_2 - m_1}{m_2 + m_1}\right) + \left(\frac{2 m_1 v_{10}}{m_1 + m_2}\right)$$

$$v_1 = v_{10}\left(\frac{m_1 - m_2}{m_1 + m_2}\right) + \left(\frac{2 m_2 v_{20}}{m_1 + m_2}\right)$$

Si $v_{20} = 0 \Rightarrow v_2 = \dfrac{2 m_1}{m_1 + m_2} v_{10}$

$$v_1 = \frac{m_1 - m_2}{m_1 + m_2} v_{10}$$

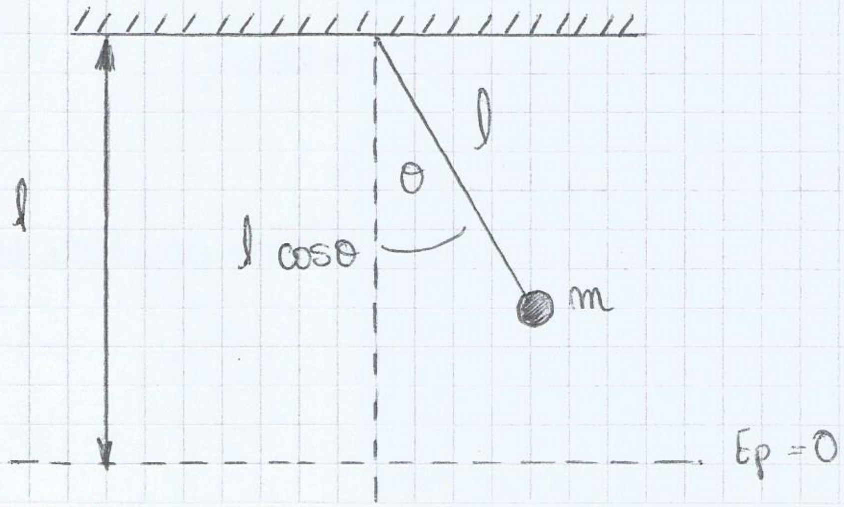
- Casos límites:

1) $m_1 = m_2 \Rightarrow v_2 = v_{10}$, $v_1 = 0$

2) $m_1 >> m_2 \Rightarrow v_2 = 2 v_{10}$, $v_1 \simeq v_{10}$ } Caso camión chocando a una pelota.

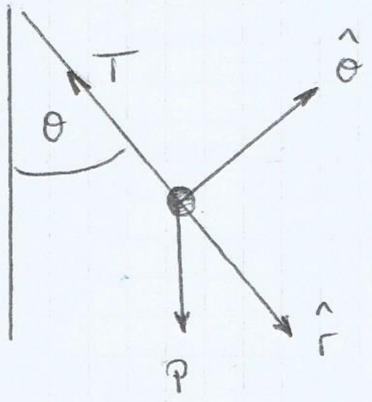
3) $m_1 << m_2 \Rightarrow v_2 \simeq 0$, $v_1 = - v_{10}$ } Caso de una pelota chocando contra una pared.

Conservación de un péndulo simple:



NOTA

1) Conservación de P



P no se conserva:

$$\frac{dP}{dt} = \sum F^{ext} = mg + T \neq 0$$

2) Conservación de L:

$$\frac{dL^{(0)}}{dt} = r \times (mg + T)$$

$r = l\hat{r}$

$T = T\hat{r}$

$mg = mg(\cos\theta\,\hat{r} - \operatorname{sen}\theta\,\hat{\theta})$

$$L^{(0)} = r \times p$$
$$L^{(0)} = l\hat{r} \times (T\hat{r} + mg\cos\theta\,\hat{r} - mg\operatorname{sen}\theta\,\hat{\theta})$$
$$L^{(0)} = l\hat{r} \times ml\dot{\theta}\,\hat{\theta}$$
$$L^{(0)} = ml^2\dot{\theta}\,\hat{z}$$

$$\frac{dL^{(0)}}{dt} = \frac{d(ml^2\dot{\theta}\,\hat{z})}{dt} = l\hat{r} \times (-T\hat{r} + mg)$$
$$ml^2\ddot{\theta}\,\hat{z} = -mgl\operatorname{sen}\theta\,\hat{z}$$
$$l\ddot{\theta} = g\operatorname{sen}\theta$$

3) Conservación de la energía mecánica:

$$E = \frac{1}{2}m(l\dot{\theta})^2 + mgl(1 - \cos\theta)$$

Si $P_\theta = ml\dot{\theta}$

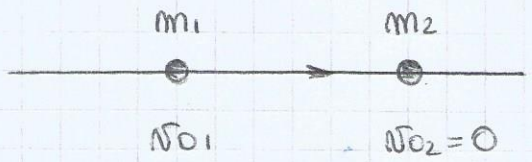
$$E = \frac{1}{2m}P_\theta^2 + mgl(1 - \cos\theta)$$

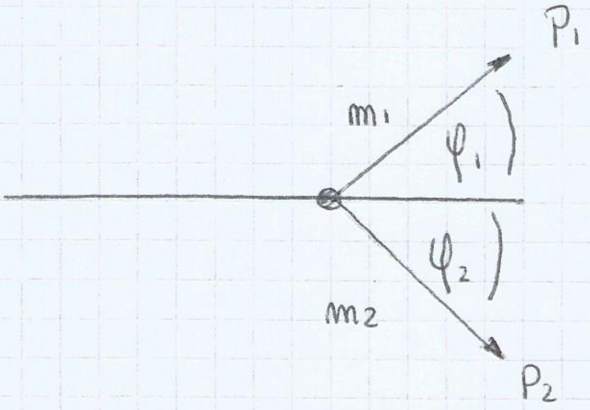
Choque plástico:

$P = cte \Rightarrow v_{cm} = cte$

$E_c = cte$

Sistema Laboratorio:





Sistema $m_1 + m_2$

1) $P = P_1 + P_2 = cte = (m_1 + m_2)\, v_{cm}$

2) $E_c = cte$

$$P = m_1 v_{10} = m_1 v_{10}\, \hat{x}$$

$$P = P\hat{x} \quad \forall t$$

$$v_{cm} = \frac{m_1 v_{10}}{m_1 + m_2}\, \hat{x} = cte$$

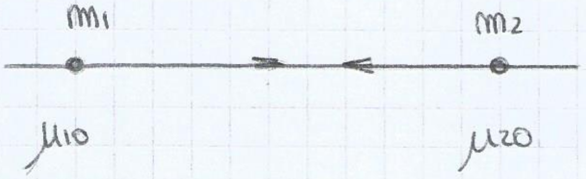
Después del choque:

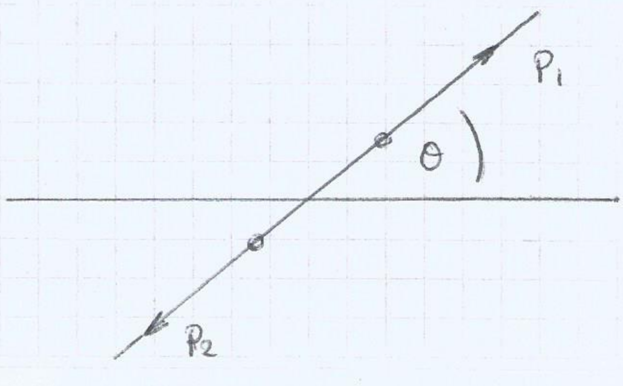
$$P = P_1 + P_2$$

$$P_1 = P_{x1}\, \hat{x} + P_{y1}\, \hat{y}$$

$$P_2 = P_{x2}\, \hat{x} + P_{y2}\, \hat{y}$$

Sistema CM:





$\mu_{10} = v_{10} - v_{cm}$

$\mu_{10} = v_{10}\,\hat{x} - \dfrac{m_1 v_{10}}{m_1 + m_2}\,\hat{x}$

Si $\alpha = \dfrac{m_1}{m_2}$

$P' = 0 \quad \forall t$

$\mu_{10} = \dfrac{m_2 v_{10}}{m_1 + m_2}\,\hat{x}$

$\mu_{10} = \dfrac{v_{10}}{1+\alpha}\,\hat{x}$

$\mu_{20} = -v_{cm} = \dfrac{-\alpha}{1+\alpha}\, v_{10}\,\hat{x}$

$$\text{II}\begin{cases} m_1\mu_{10} + m_2\mu_{20} = 0 & \Rightarrow \mu_{20} = -\alpha\,\mu_{10} \\ m_1\mu_1 + m_2\mu_2 = 0 & \Rightarrow \mu_2 = -\alpha\,\mu_1 \end{cases}$$

$$E_c = cte = \frac{1}{2} m_1 \mu_{10}^2 + \frac{1}{2} m_2 \mu_{20}^2 = \frac{1}{2} m_1 \mu_1^2 + \frac{1}{2} m_2 \mu_2^2$$

$$\alpha\,\mu_{10}^2 + \alpha\,\mu_{20}^2 = \alpha\,\mu_1^2 + \alpha\,\mu_2^2$$

- Usando la ecuación II:

$$(\alpha + \alpha^2)\,\mu_{10}^2 = (\alpha + \alpha^2)\,\mu_1^2 \qquad \left.\begin{aligned} \mu_{10} &= \mu_1 \\ \mu_{20} &= \mu_2 \end{aligned}\right\} \text{en módulo}$$

$$\mu_1 = \frac{1}{1+\alpha} v_{10} (\cos\theta\,\hat{x} + \operatorname{sen}\theta\,\hat{y}) = \frac{v_{cm}}{\alpha} (\cos\theta\,\hat{x} + \operatorname{sen}\theta\,\hat{y})$$

$$\mu_2 = \frac{-\alpha}{1+\alpha} v_{10} (\cos\theta\,\hat{x} + \operatorname{sen}\theta\,\hat{y}) = -v_{cm} (\cos\theta\,\hat{x} + \operatorname{sen}\theta\,\hat{y})$$

$$v_1 = \mu_1 + v_{cm} = (\mu_1 \cos\theta + v_{cm})\,\hat{x} + (\mu_1 \operatorname{sen}\theta)\,\hat{y}$$

$$v_2 = \mu_2 + v_{cm} = (\mu_2 \cos\theta + v_{cm})\,\hat{x} + (\mu_2 \operatorname{sen}\theta)\,\hat{y}$$

NOTA

$$\operatorname{tg}\varphi_1 = \frac{v_{1y}}{v_{1x}} = \frac{\mu_1 \operatorname{sen}\theta}{\mu_1 \cos\theta + v_{cm}} = \frac{\operatorname{sen}\theta}{\operatorname{sen}\theta + \frac{v_{cm}}{\mu_1}} = \frac{\operatorname{sen}\theta}{\cos\theta + \alpha}$$

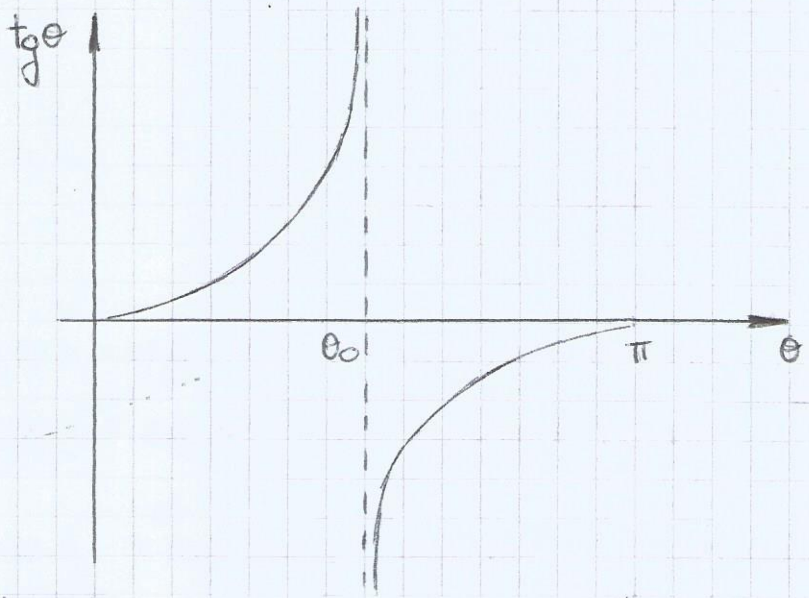
$$\operatorname{tg}\varphi_2 = \frac{v_{2y}}{v_{2x}} = \frac{\mu_2 \operatorname{sen}\theta}{\mu_2 \cos\theta + v_{cm}} = \frac{\operatorname{sen}\theta}{\cos\theta + \frac{v_{cm}}{\mu_2}} = \frac{\operatorname{sen}\theta}{\cos\theta - 1}$$

a) $\alpha < 1$, $m_1 < m_2$

Si $\cos\theta_0 = -\alpha$

$-\infty \leq \operatorname{tg}\varphi_1 \leq \pi$

$0 \leq \varphi_1 \leq \pi$

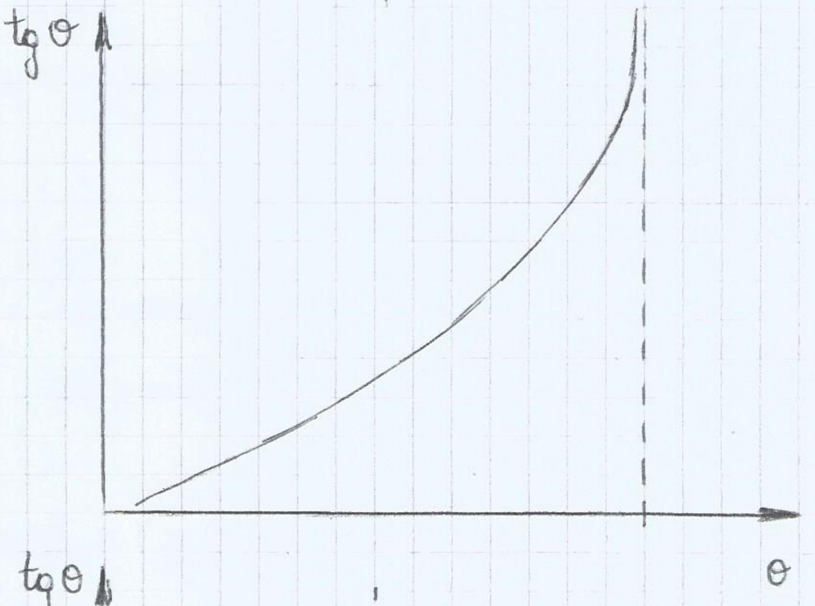


b) $\alpha = 1$, $m_1 = m_2$

Si $\cos\theta \geq -1$

$0 \leq \operatorname{tg}\varphi_1 \leq \infty$

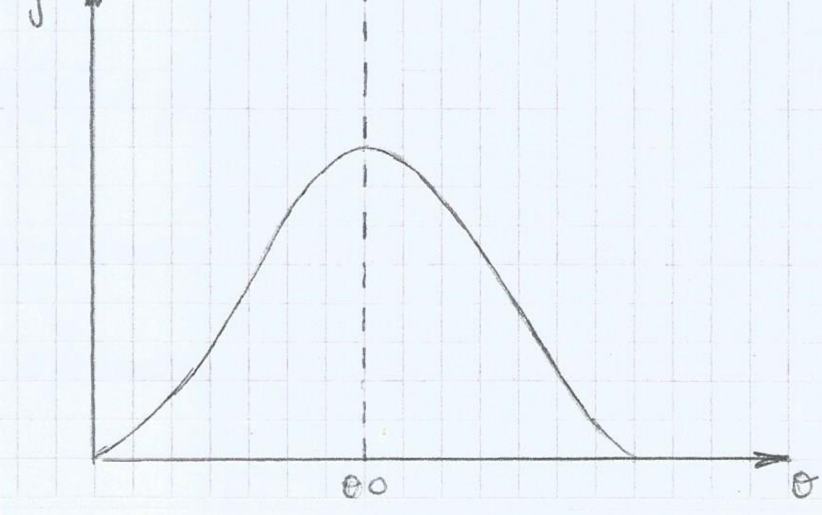
$0 \leq \varphi_1 \leq \frac{\pi}{2}$



c) $\alpha > 1$, $m_1 > m_2$

$0 \leq \operatorname{tg}\varphi_1 < \infty$

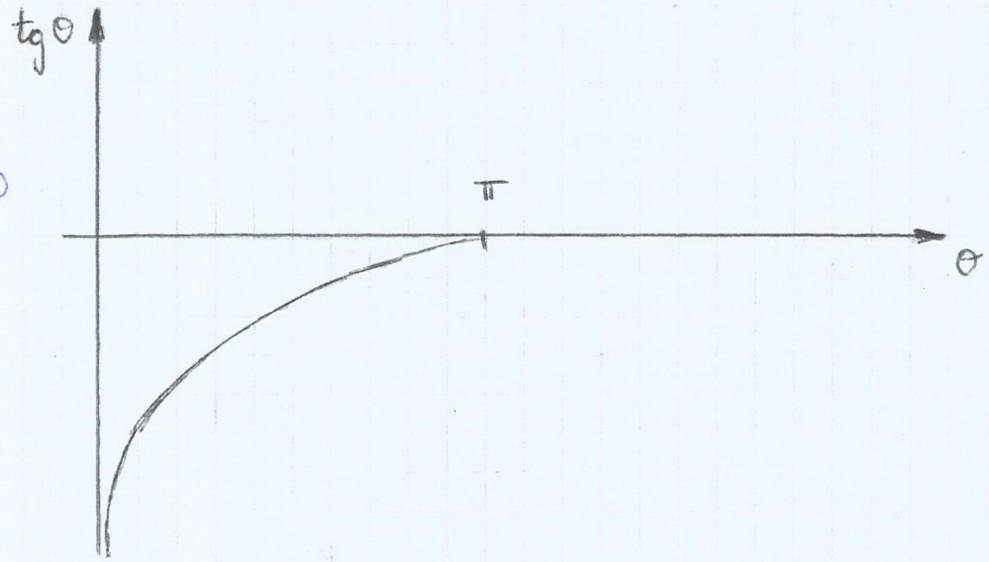
$0 \leq \varphi_1 < \frac{\pi}{2}$



NOTA

Analizo φ_2

$-\infty \leq tg\,\varphi_2 \leq 0$



Teoremas de conservación:

$$P = \sum_{i=1}^{N} P_i = M\, v_{CM} \quad , \quad F^{ext} = 0 \;,\; P = cte$$

$$F = \frac{dP}{dt} = \sum_{i=1}^{N} \frac{dP}{dt}$$

$$= \sum_{i=1}^{N} F_i^{ext} + \sum_{i=1}^{N} \sum_{j \neq 1}^{N} f_{ji}$$

$$R_{cm} = \frac{\sum^{N} m_i\, r_i}{M}$$

$$E_m = E_c + E_p$$

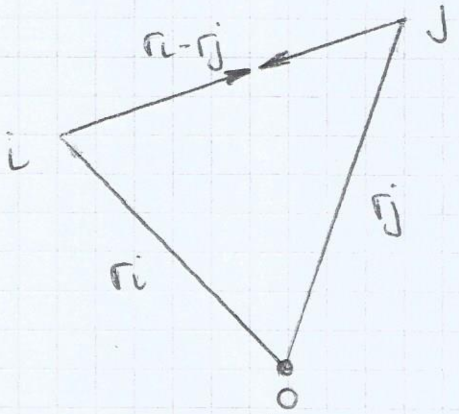
$$E_m = \sum E_{mi} \qquad dE_m = \sum_{i=1}^{N} dE_{mi} = \sum_{i=1}^{N} dW_i^{NC} = dW^{NC}$$

$$dW^{NC} = dW^{NC,ext} + dW^{NC,int}$$

$$dW^{NC} = -\sum F_i^{ext} dr_i + \sum_{i=1}^{N} \sum_{j \neq 1} f_{ji}\, dr_i$$

$$f_{ji}\,dr_i + f_{ij}\,dr_j = f_{ji}(dr_i - dr_j)$$
$$= f_{ji}\,d\underbrace{(r_i - r_j)}_{cte}$$

$f_{ij} \parallel r_i - r_j$



$$E_M = cte \iff W^{NC} = W^{NC,ext} + W^{NC,int} = 0$$

$$L_o = \sum_{i=1}^{N} L_{oi} = \sum_{i=1}^{N} r_i \times p_i$$ Si O está quieto o es el sistema CM $\Rightarrow$ $M_o = 0 \iff L_o = cte$

$F = F_i^{ext}\,;\ f_{ij}$

$$F = \sum_{i=1}^{N} F_i^{ext} + \sum_{i=1}^{N}\sum_{j\neq i} f_{ji}$$

$$\frac{dL_o}{dt} = \sum_{i=1}^{N} \frac{dL_{oi}}{dt} = \sum_{i=1}^{N} r_i \times F$$

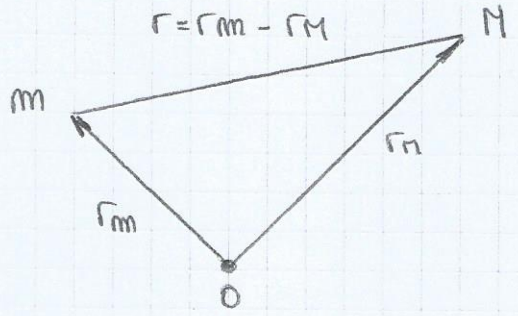
$$\frac{dL_o}{dt} = \sum_{i=1}^{N} r_i \times F_i^{ext} + \underbrace{\sum_{i=1}^{N} r_i \times \sum_{j\neq i} f_{ji}}$$

$r_i \times f_{ji} + r_j \times f_{ij}$ si $f_{ij} \parallel r_i - r_j$

$(r_i - r_j) \times f_{ji} = 0$

$$L_o = cte \iff M_o^{ext} = 0$$

Sistema de 2 cuerpos:



$\vec{F} = F(r)\,\hat{r}$

$P = cte \quad P = P_m + P_M \Rightarrow V_{cm} = cte$

$L_0 = cte$

(R_{cm} se mueve con MRU)

$$R_{cm} = \frac{m\, r_m + M\, r_M}{m + M} \qquad F = m\,\ddot{r}_m = m\, a_m$$

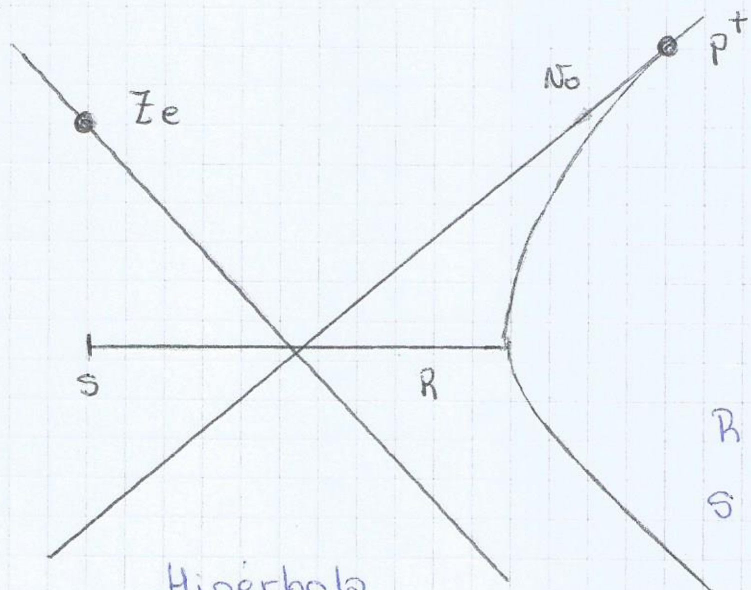
$$r = r_m - r_M \qquad -F = M\,\ddot{r}_M = M\, a_M$$

$$\ddot{R}_{cm} = \frac{m\,\ddot{r}_{cm} + M\,\ddot{r}_M}{m + M} = \frac{F - F}{m + M} = 0 \Rightarrow R_{cm}(t) = R_{cm}(0) + V_{cm}\, t$$

$$\ddot{r} = \ddot{r}_m - \ddot{r}_M = \frac{F}{m} + \frac{F}{M} = F\left(\frac{1}{m} + \frac{1}{M}\right)$$

Llamamos $\mu = \dfrac{m\,M}{m + M}$ (masa reducida).

Dispersión de un protón por un núcleo:



Hipérbola

$F_{elec} = \dfrac{K q_1 q_2}{r^2}\,\hat{r}$

$E_p = \dfrac{K q_1 q_2}{r}$

$E_M = cte \quad L_{cm} = cte$

R máxima aproximación

S parámetro de impacto